



TECHNICAL PROFESSIONAL

ICT - COMPUTER PROGRAMMING, NET

Grade 12

**Three-Term Budget Of Work (BOW)
for Learning Competencies**

TECHNICAL PROFESSIONAL
ICT - ICT SUPPORT AND COMPUTER PROGRAMMING TECHNOLOGIES
Grade 12

Three-Term Budget Of Work
For Learning Competencies

Please note that each TechPro elective is completed within one term if delivered at Grade 12.

Term Plan	
Elective	Computer Programming, Net

Week	Learning Competency
*	• Discuss the HTML fundamentals. • Use paragraph formatting elements. • Apply list attributes and values in web page designs. • Create HTML documents with multimedia elements. • Use table elements, attributes, and values in designing web pages. • Create hyperlinks between web pages. • Use frames in designing web pages. • Create HTML form web pages and provide fields where user can enter data. • Perform installation of Integrated Development Environment (IDE) application. • Discuss Cascading Style Sheet (CSS) fundamentals. • Apply the different CSS Selectors.
*	• Apply CSS text, font, and color in designing web pages.

Week	Learning Competency
	• Apply CSS gradient and opacity. • Create layouts of web pages using the CSS box model. • Use the CSS list and link properties and values in creating web pages. • Utilizes CSS Z-index, position, float properties and values in designing web pages. • Create web pages with a CSS image gallery. • Apply CSS pseudo class and elements. • Apply CSS flexbox layout and media queries in web designs. • Create responsive websites using CSS Responsive Web Development (RWD) Layouts. • Discuss the fundamentals of JavaScript. • Demonstrate the structures of statements, algorithm, flowchart, and pseudocode. • Apply variables, data types, and functions in creating web applications. • Use JavaScript operators in problem solving. • Use event handlers and event listeners in creating web applications. • Use conditional and switch statements in creating web applications. • Apply loops in creating a program using while, do while, and for loops. • Use JavaScript arrays in creating a program. • Create objects to display the set of properties and values. • Apply Document Object Model (DOM) nodes in creating a web document.
*	• Discuss the C# fundamentals. • Use C# data types, variables, strings, and constants in creating a web application. • Apply C# operators and decision making in problem solving. • Use the C# iteration statements in creating a web application. • Apply C# classes, methods and structure in creating a web application. • Create web application programs that contains C# arrays.

Week	Learning Competency
	• Apply the C# Object-Oriented Programming (OOP) approach in creating a web application. • Utilize the C# file input/output in creating, deleting, and reading web applications. • Discuss the ASP.NET Model View Controller (MVC). • Discuss the ASP.NET MVC environment and life cycle. • Perform configurations of the routing process in directing an http request in an ASP.NET MVC project using controllers and actions. • Use filters, selectors, and views in creating projects. • Apply the ASP.NET MVC - data model. • Use ASP.NET MVC – helpers in creating realistic HTML controls. • Create bridges between the http request and the action methods (Model Binding). • Discuss the framework of ASP.NET MVC – database and adding validation to a model. • Execute security features in a web application. • Create ASP.NET MVC - web application program interface (API). • Convert website to mobile version. • Compute the service costs for web development, web hosting.

* Teachers may assign learning competencies per week depending on the needs of the learners.